

SARA BAKHTIARI

PhD Candidate in Mechanical Engineering | Computational Mechanics | Finite Element Modeling | Scientific Computing

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Nonlinear FEA · Constitutive Modeling · Cardiac Mechanics · HPC · Python · FEniCS/DOLFIN · MPI/PETSc

PROFESSIONAL SUMMARY

PhD candidate in Mechanical Engineering at the University of Kentucky specializing in computational mechanics and biomechanics. Develops nonlinear finite-element models of the left ventricle that couple passive constitutive behavior, active contraction, circulation, and remodeling. Experienced in Python, FEniCS/DOLFIN, MPI/PETSc, SLURM-based HPC, model validation, and scientific post-processing, with additional experience in machine learning for engineering data.

EDUCATION

University of Kentucky — Lexington, KY

Spring 2024–Present

PhD in Mechanical Engineering, Direct PhD Program · PhD Candidate since April 2026 · GPA: 3.91/4.00 · Advisor: Jonathan F. Wenk, PhD

Iran University of Science and Technology (IUST) — Tehran, Iran

2016–2021

B.Sc. in Mechanical Engineering

RESEARCH EXPERIENCE

Graduate Research Assistant — Computational Cardiac Mechanics

Spring 2024–Present

University of Kentucky · Advisor: Jonathan F. Wenk, PhD

- Develop and extend MyoFE_H, a multiphysics finite-element framework coupling nonlinear left-ventricular mechanics, spatially distributed half-sarcomere kinetics, active contraction, and lumped-parameter circulation.
- Implement and evaluate anisotropic passive myocardial constitutive laws, including Guccione- and Holzapfel-Ogden-based formulations, to quantify effects on stress, pressure-volume behavior, deformation, and growth/remodeling predictions.
- Developed computational studies of myocardial fiber disarray and quantified its effects on ventricular mechanics, mechanical efficiency, energetics, and baroreflex compensation.
- Currently developing an agent-based fibrosis model coupled with finite-element cardiac mechanics to study spatially evolving fibrosis and tissue-level mechanical adaptation.
- Run distributed simulations using FEniCS/DOLFIN, Python, MPI, PETSc, SLURM, and Linux HPC environments; perform nonlinear Newton solves and quadrature-point calculations.
- Develop validation and post-processing workflows for biaxial/shear material tests, passive ventricular inflation, stress/strain fields, pressure-volume loops, energetics, and ParaView visualization.

Machine Learning for Engineering Data

Additional research project

- Applied pretrained and fine-tuned vision-language models to electrospinning videos to extract high-dimensional representations for fiber-diameter prediction.
- Evaluated embedding repeatability, bottleneck representations, feature variability, regression performance, and repeated cross-validation.

PUBLICATIONS & RESEARCH OUTPUTS

Islam, R., Bakhtiari, S., Anugrah, G. G., Mehri, M., & Wenk, J. F. “Advancements in finite element modeling of myocardial infarction: a review.” *Biomechanics and Modeling in Mechanobiology*, 25(4), 2026. DOI: 10.1007/s10237-026-02107-0

Manuscripts Under Review

- Bakhtiari, S., Mehri, M., & Wenk, J. F. “Choice of Passive Myocardial Constitutive Law Drives Divergent Predictions of Left Ventricular Remodeling in Volume-Overload Conditions.” Manuscript under review.
- Bakhtiari, S., Mehri, M., & Wenk, J. F. “Effect of Myocardial Fiber Disarray on Left Ventricular Mechanical Efficiency and Baroreflex Compensation.” Manuscript under review.

Selected Conference Presentation

Bakhtiari, S., Mehri, M., & Wenk, J. F. “Choice of Passive Myocardial Constitutive Law Drives Differences in LV Mechanics and Growth Under Disease Conditions.” Poster, Biomedical Engineering Society (BMES) Annual Meeting, San Diego, CA, October 2025.

TEACHING EXPERIENCE

Teaching Assistant — Statics, University of Kentucky

Fall 2026

Supported undergraduate instruction and problem solving in equilibrium, force/moment analysis, trusses, and frames.

Teaching Assistant — Statics, Iran University of Science and Technology

2021

TECHNICAL SKILLS

Computational Mechanics: Nonlinear FEA, continuum mechanics, constitutive modeling, anisotropic soft-tissue mechanics, mixed incompressible formulations, Newton-Raphson methods

Scientific Computing: Python, MATLAB, NumPy, SciPy, Pandas, FEniCS/DOLFIN, UFL

HPC: MPI, mpi4py, PETSc, SLURM, Linux, distributed-memory computing, Singularity/container workflows

Cardiovascular Modeling: Cardiac mechanics, active contraction, pressure-volume analysis, lumped circulation, growth/remodeling, fibrosis, fiber architecture, energetics

Visualization: ParaView, Matplotlib, HDF5, XDMF, CSV, NumPy arrays